Mirza Naeem Beg

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SUMMARY

I am a recent Computer Science and Engineering graduate from *AUST*. I enjoy solving problems where language, vision, and data come together, and that's what got me interested in AI/ML. My background includes deep learning research, full-stack web development, and experience working with SQL and data. I am currently deepening my knowledge of deep learning using PyTorch, Keras, and TensorFlow while building my AI engineering skills step by step, following a self-designed learning plan to prepare for an AI engineering role. I am looking for a junior or trainee role in software engineering or AI engineering where I can apply my knowledge, collaborate with experienced engineers, and continue developing my skills professionally.

EDUCATION

Ahsanullah University of Science and Technology (AUST) **Dhaka, Bangladesh**
B.Sc. in Computer Science and Engineering | CGPA: 3.13 / 4.00 **July 2026**
Relevant coursework: Data Structures, Algorithms, OOP, Database Systems, Intro. to AI, Pattern Recognition & ML, Soft Computing, Data Mining, Digital Image Processing, Software Development I–V, Mathematics I-IV

Mymensingh Government College **Mymensingh Board**
Higher Secondary Certificate (HSC) | Science | GPA: 5.00 / 5.00 **2020**

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, C/C++
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, Hugging Face
- **AI & Machine Learning:** CNNs, RNNs/BiLSTM, Computer Vision, NLP, Transformers
- **Databases & SQL:** PostgreSQL, MySQL, Microsoft SQL Server — Joins, Subqueries & CTEs, Window Functions, Indexing, Triggers, Transactions, NULL Handling, Soft Deletion, Schema Design
- **Web Development:** HTML, CSS, Bootstrap, TailwindCSS, ReactJS, .NET Core
- **Tools:** Git & GitHub, Shell, Jupyter Notebook, Google Colab, Kaggle, VS Code, Power BI

UNDERGRADUATE THESIS

BanglaGuard: A Multimodal Deep Learning Framework for Audio-Visual Detection of Inappropriate Content in Bangla Children's Cartoon Videos **Defended June 2026**
Supervisor: Prof. Dr. Md. Shahriar Mahbub *Poster Presented June 2026*

- Built a multimodal pipeline combining EfficientNet-based CNNs for visual features, a spectrogram-based CNN for Bangla audio classification, and a BiLSTM for temporal modeling.
- Achieved 87.02% accuracy on a purpose-built 871-clip dataset; under review for conference publication.

ACADEMIC PROJECTS

Digital Preservation of Culinary Heritage: Bangladeshi Dessert Classification & Recipe Generation | *Soft Computing Lab Term Assignment* **Feb 2025 – Jul 2025**

- Built a multimodal deep learning system using EfficientNet-B0 and GPT-2 on a curated *dataset* of 4,118 images and 81 traditional recipes to digitally preserve Bangladeshi culinary heritage.

BechaKena.com: Full-Stack E-Commerce & Auction Platform **Aug 2024 – Nov 2024**
Software Development IV Course Project | youtu.be/JnbI_pToNd4

- Designed the relational database schema in MySQL and wrote 20+ complex SQL queries (joins, subqueries, aggregation) to support real-time bidding, transactions, and user management with zero data loss.
- Built with HTML, CSS, Bootstrap and PHP. Structured the platform into separate modules such as add-to-cart, user profile, and checkout, each handling its own logic independently, so the codebase stayed easy to extend, test, and debug as features grew. *github.com/mirzanaeembeg/CSE3100-SD-IV*

Railway Ticket Management System: Desktop Application **2023**
Software Development II Course Project

- Built a JavaFX/MySQL ticket booking system using SQL queries, Scene Builder, and XAMPP.

CERTIFICATIONS

- *Associate AI Engineer for Data Scientists,* DataCamp
- *Associate AI Engineer for Developer,* DataCamp
- *CS50's Introduction to Databases with SQL,* Harvard University

CORE COMPETENCIES

• Problem-Solving • Object-Oriented Development • Database Design • Mathematics, Probability & Statistics • Adaptability • Quality Assurance • Attention to Detail • Reliability • Cross-Functional Collaboration